Erdős Conjecture on Arithmetic Progressions (Problem 77)

Abstract

We investigate the Erdős Conjecture on Arithmetic Progressions, formalizing the necessary axiomatic definitions, reviewing the bounds associated with k-term arithmetic progressions, and establishing fundamental density lemmas. A formal architecture for Lean 4 is also provided.

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1 Analysis and Decomposition

The Erdős conjecture on arithmetic progressions states that if the sum of the reciprocals of the members of a set A of positive integers diverges, then A contains arbitrarily long arithmetic progressions.

Definition 1.1. *A set $A \subset \mathbb{N}_{>0}$ is termed large if:*

$$\sum_{n \in A} \frac{1}{n} = \infty$$

Definition 1.2. *For $k \in \mathbb{N}_{\geq 3}$, a k-term arithmetic progression in A is a sequence of the form $\{a, a+d, a+2d, \ldots, a+(k-1)d\} \subset A$ for some $a \in \mathbb{N}$ and $d \in \mathbb{N}_{>0}$.*

2 Contextual Literature Research

This problem is listed as Problem 16 in Bloom's list of Erdős' problems. It closely relates to bounds on $r_k(N)$, the maximum cardinality of a subset of $\{1, 2, \ldots, N\}$ that contains no k-term arithmetic progression.

3 Proof Strategy & Isolation of Lemmas

Our strategy involves establishing that if $A(N) = |A \cap \{1, \dots, N\}|$ is sufficiently bounded, the sum of reciprocals converges. We divide this into two main lemmas:

- **Lemma 1**: An exact summation formula linking the reciprocal sum to $A(n)/n(n+1)$.
- **Lemma 2**: A convergence result using the integral test given a specific bound on $A(n)$.

4 Informal Proof

Let $r_k(N)$ denote the maximum cardinality of a subset of $\{1, 2, \dots, N\}$ that contains no k-term arithmetic progression. If A contains no k-term arithmetic progressions, then for every integer $N \geq 1$, we have $|A \cap \{1, \dots, N\}| \leq r_k(N)$.

Lemma 4.1. *Let $A(N) = |A \cap \{1, \dots, N\}|$. For any positive integer M, the finite sum of reciprocals satisfies:*

$$\sum_{\substack{n \in A \\ n \leq M}} \frac{1}{n} = \sum_{n=1}^{M} \frac{A(n)}{n(n+1)} + \frac{A(M)}{M+1}$$

Proof. By definition, the indicator function of A, denoted $1_A(n)$, allows us to write $A(n) - A(n-1) = 1_A(n)$, with $A(0) = 0$. We expand the sum:

$$\begin{aligned}
\sum_{\substack{n \in A \\ n \leq M}} \frac{1}{n} &= \sum_{n=1}^{M} \frac{1_A(n)}{n} \\
&= \sum_{n=1}^{M} \frac{A(n) - A(n-1)}{n} \\
&= \frac{A(1) - A(0)}{1} + \frac{A(2) - A(1)}{2} + \cdots + \frac{A(M) - A(M-1)}{M} \\
&= A(1)\left(1 - \frac{1}{2}\right) + A(2)\left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + A(M-1)\left(\frac{1}{M-1} - \frac{1}{M}\right) + \frac{A(M)}{M} \\
&= \sum_{n=1}^{M-1} A(n)\left(\frac{1}{n} - \frac{1}{n+1}\right) + \frac{A(M)}{M} \\
&= \sum_{n=1}^{M-1} A(n)\left(\frac{n+1-n}{n(n+1)}\right) + \frac{A(M)}{M} \\
&= \sum_{n=1}^{M-1} \frac{A(n)}{n(n+1)} + \frac{A(M)}{M}
\end{aligned}$$

Notice that $\frac{A(M)}{M} = \frac{A(M)}{M(M+1)} + \frac{A(M)}{M+1}$, so the identity holds identically for M. $\square$

Lemma 4.2. *If there exist constants $C > 0$ and $\epsilon > 0$ such that $A(n) \leq C\frac{n}{(\log n)^{1+\epsilon}}$ for all $n \geq 2$, then $\sum_{n\in A} \frac{1}{n} < \infty$.*

Proof. Applying the previous lemma, as $M \to \infty$, since $A(M) \leq M$, $\frac{A(M)}{M+1} \leq 1$. The infinite sum $\sum_{n\in A} \frac{1}{n}$ converges if and only if $\sum_{n=1}^{\infty} \frac{A(n)}{n(n+1)}$ converges. Using the bound on $A(n)$, for $n \geq 2$:

$$\frac{A(n)}{n(n+1)} \leq \frac{C\frac{n}{(\log n)^{1+\epsilon}}}{n(n+1)} = \frac{C}{(n+1)(\log n)^{1+\epsilon}}$$

Since $\frac{1}{n+1} < \frac{1}{n}$, we have:

$$\frac{A(n)}{n(n+1)} < \frac{C}{n(\log n)^{1+\epsilon}}$$

The integral test for convergence states that $\sum_{n=2}^{\infty} f(n)$ converges if $\int_2^{\infty} f(x)dx < \infty$ for a positive monotonically decreasing function. Let $f(x) = \frac{1}{x(\log x)^{1+\epsilon}}$. We compute the integral using substitution $u = \log x, du = \frac{1}{x}dx$:

$$\int_2^{\infty} \frac{1}{x(\log x)^{1+\epsilon}}dx = \int_{\log 2}^{\infty} \frac{1}{u^{1+\epsilon}}du = \left[\frac{-1}{\epsilon u^{\epsilon}}\right]_{\log 2}^{\infty} = \frac{1}{\epsilon(\log 2)^{\epsilon}}$$

Since the integral evaluates to a finite constant, the sum $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^{1+\epsilon}}$ converges, implying $\sum_{n\in A} \frac{1}{n}$ converges. $\square$

5 Architecture for Autoformalization

```
import Mathlib.Data.Real.Basic
import Mathlib.Data.Finset.Basic
import Mathlib.Analysis.SpecialFunctions.Log.Basic

def has_k_AP (A : Set Nat) (k : Nat) : Prop :=
  Exists (fun a => Exists (fun d => d > 0 /\ \forall  i < k, (a + i * d) \in  A))

theorem erdos_ap_conjecture (A : Set Nat) :
  (tsum (fun n : A => (1 : \mathbb{R}) / (n : \mathbb{R})) = \top ) -> \forall  k >=
```